

Two-Way-ANOVA

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TWO-WAY ANOVA

Data Loading

```
data <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/Two_Way_ANOVA/
data$Fertilizer <- as.factor(data$Fertilizer)
data$Irrigation <- as.factor(data$Irrigation)
```

Model Building

```
model <- aov(Yield ~ Fertilizer * Irrigation, data = data)
```

Assumption Checking

a. Normality of residuals

```
shapiro.test(residuals(model))
```

Shapiro-Wilk normality test

data: residuals(model)

W = 0.86093, p-value = 0.003511

$p > 0.05 \rightarrow$ residuals are normally distributed

b. Homogeneity of variance

```
library(car)
```

Warning: package 'car' was built under R version 4.4.3

Loading required package: carData

```
leveneTest(Yield ~ Fertilizer * Irrigation, data = data)
```

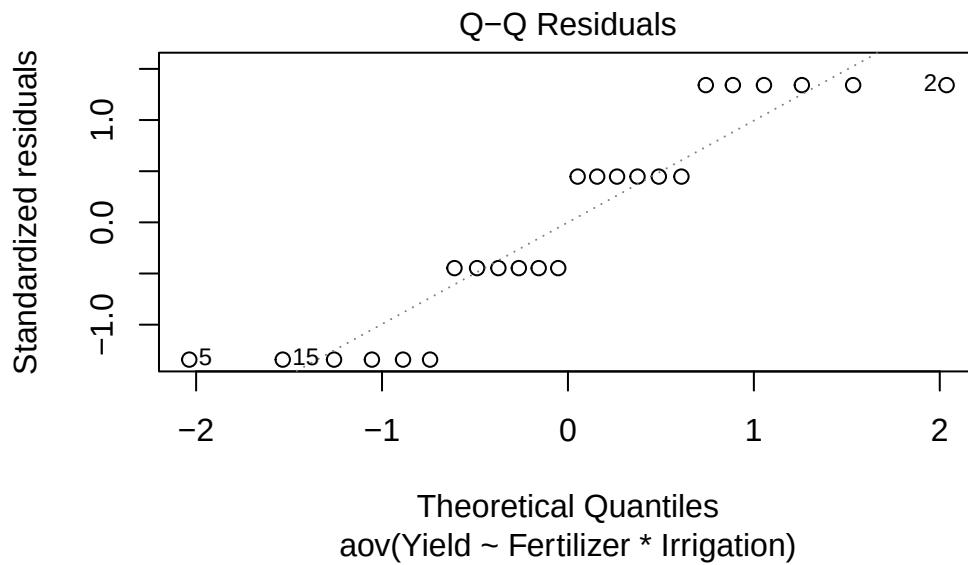
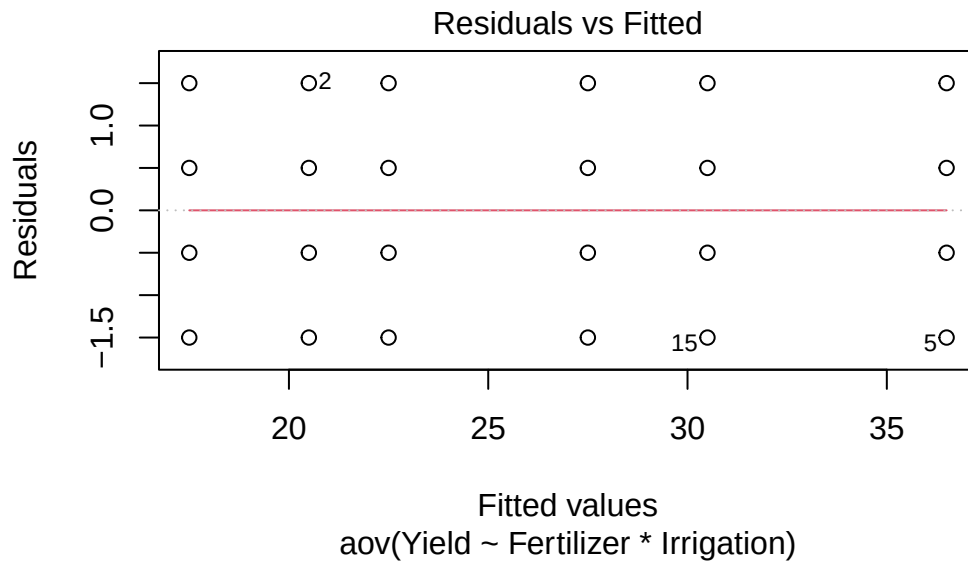
Levene's Test for Homogeneity of Variance (center = median)

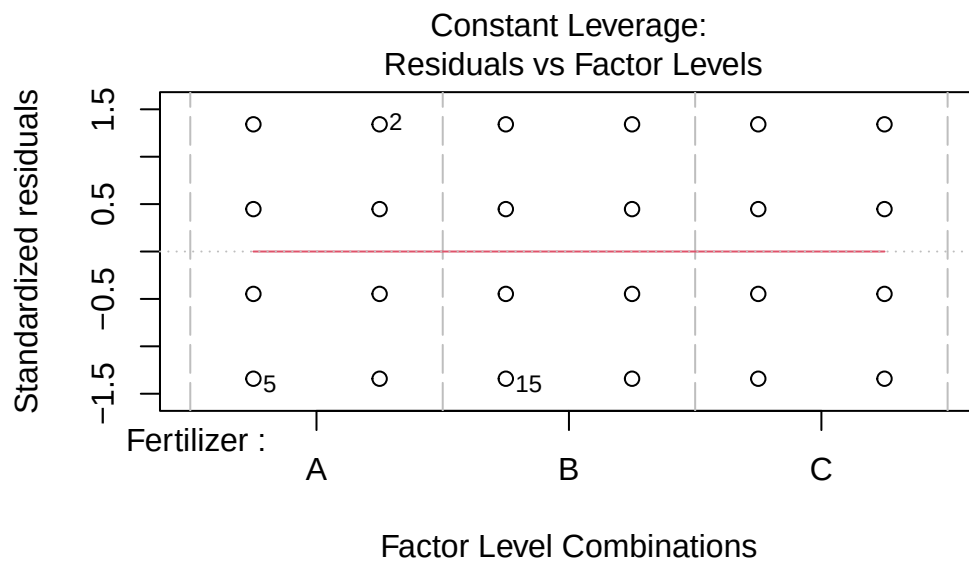
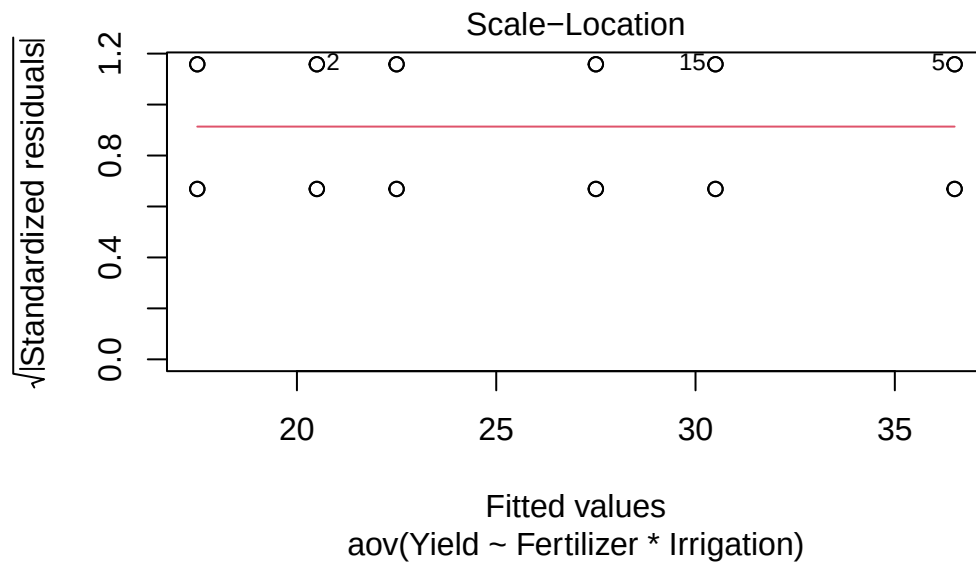
	Df	F value	Pr(>F)
group	5	0	1
	18		

$p > 0.05 \rightarrow$ equal variance assumption satisfied

c. Residual diagnostics

```
plot(model)
```





No strong pattern $\rightarrow$ model is valid

Perform ANOVA

```
summary(model)
```

```
              Df Sum Sq Mean Sq F value    Pr(>F)
Fertilizer      2  409.3   204.7    122.8 3.23e-11 ***
Irrigation      1  384.0   384.0    230.4 1.06e-11 ***
Fertilizer:Irrigation  2  196.0    98.0    58.8 1.28e-08 ***
Residuals     18   30.0     1.7
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Testing 3 hypotheses:

- Fertilizer effect (main effect)
- Irrigation effect (main effect)
- Fertilizer $\times$ Irrigation interaction (joint effect)

Interpretation of ANOVA Table

Main Effects-

Fertilizer effect : $p < 0.05 \rightarrow$ Fertilizer types are NOT equal. Means at least one fertilizer changes yield

Irrigation effect : $p < 0.05 \rightarrow$ Irrigation levels significantly affect yield

CRITICAL WARNING (Most beginners miss this)

If interaction is significant, main effects alone are NOT reliable because:

Effect of fertilizer changes depending on irrigation So “average fertilizer effect” becomes misleading

Interaction Effect (Most Important Part)

Fertilizer $\times$ Irrigation: $p < 0.05 \rightarrow$ Interaction exists

Correct meaning:

The effect of Fertilizer on Yield depends on Irrigation level (and vice versa)

NOT:

- “They are related”
- “Both are important”

Proper Statistical Decision Rule

Case 1: Interaction NOT significant

Interpret fertilizer main effect and Irrigation main effect separately

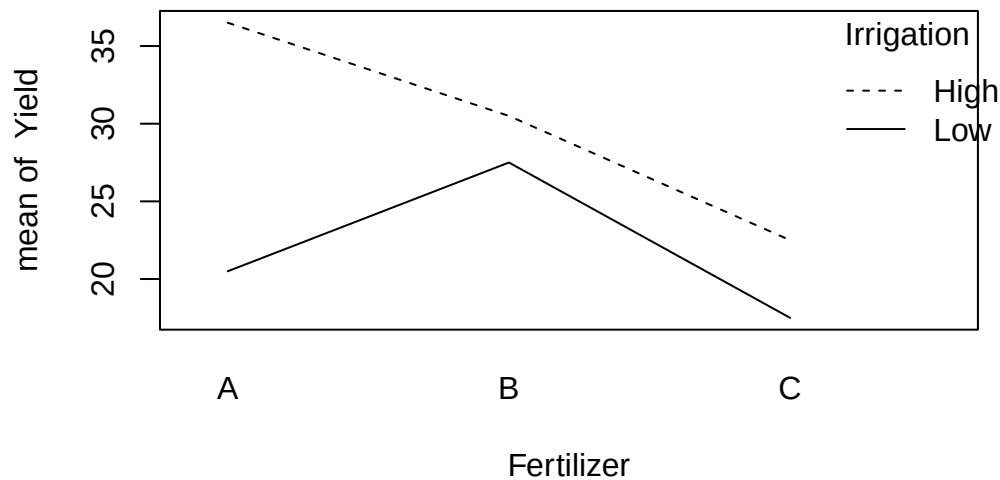
Case 2: Interaction significant

IGNORE main effects

Focus on: Interaction plots, Post-hoc within each irrigation level

Interaction Plot

```
with(data, interaction.plot(Fertilizer, Irrigation, Yield))
```



Interpret Interaction Plot:

Step 1: Check line crossing

- Lines crossing → strong interaction
- Lines parallel → no interaction

Step 2: Interpret patterns scientifically

Example interpretation:

Fertilizer A:

High irrigation → highest yield

Low irrigation → sharp drop

That's indicate **HIGH water dependency**

Fertilizer B:

Stable across irrigation levels → drought-tolerant performance

Fertilizer C:

Low performance everywhere → generally inefficient

Mean table

```
aggregate(Yield ~ Fertilizer + Irrigation, data = data, mean)
```

	Fertilizer	Irrigation	Yield
1	A	High	36.5
2	B	High	30.5
3	C	High	22.5
4	A	Low	20.5
5	B	Low	27.5
6	C	Low	17.5

POST HOC

Global turkey

TukeyHSD(model)

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = Yield ~ Fertilizer * Irrigation, data = data)

\$Fertilizer

	diff	lwr	upr	p adj
B-A	0.5	-1.147414	2.147414	0.722937
C-A	-8.5	-10.147414	-6.852586	0.000000
C-B	-9.0	-10.647414	-7.352586	0.000000

\$Irrigation

	diff	lwr	upr	p adj
Low-High	-8	-9.107283	-6.892717	0

\$`Fertilizer:Irrigation`

	diff	lwr	upr	p adj
B:High-A:High	-6	-8.901136	-3.09886432	0.0000456
C:High-A:High	-14	-16.901136	-11.09886432	0.0000000
A:Low-A:High	-16	-18.901136	-13.09886432	0.0000000
B:Low-A:High	-9	-11.901136	-6.09886432	0.0000001
C:Low-A:High	-19	-21.901136	-16.09886432	0.0000000
C:High-B:High	-8	-10.901136	-5.09886432	0.0000009
A:Low-B:High	-10	-12.901136	-7.09886432	0.0000000
B:Low-B:High	-3	-5.901136	-0.09886432	0.0403048
C:Low-B:High	-13	-15.901136	-10.09886432	0.0000000
A:Low-C:High	-2	-4.901136	0.90113568	0.2889127
B:Low-C:High	5	2.098864	7.90113568	0.0004115
C:Low-C:High	-5	-7.901136	-2.09886432	0.0004115
B:Low-A:Low	7	4.098864	9.90113568	0.0000058
C:Low-A:Low	-3	-5.901136	-0.09886432	0.0403048
C:Low-B:Low	-10	-12.901136	-7.09886432	0.0000000

Problem with Global Tukey

This compares all combinations together, which:

- mixes irrigation + fertilizer effects
- hides the actual interaction structure

Result = confusing + misleading conclusions. That's why this is technically correct but scientifically wrong.

Correct Approach: Simple Effects Analysis

1. Effect of Fertilizer at each Irrigation level

```
#Effects of fertilizer when irrigation low
model_low <- aov(Yield ~ Fertilizer, data = subset(data, Irrigation == "Low"))
TukeyHSD(model_low)
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = Yield ~ Fertilizer, data = subset(data, Irrigation == "Low"))
```

```
$Fertilizer
      diff      lwr      upr      p adj
B-A      7  4.451259  9.5487408 0.0000823
C-A     -3 -5.548741 -0.4512592 0.0231730
C-B    -10 -12.548741 -7.4512592 0.0000045
```

```
#Effects of fertilizer when irrigation high
model_high <- aov(Yield ~ Fertilizer, data = subset(data, Irrigation == "High"))
TukeyHSD(model_high)
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = Yield ~ Fertilizer, data = subset(data, Irrigation == "High"))
```

```
$Fertilizer
      diff      lwr      upr      p adj
B-A     -6 -8.548741 -3.451259 0.0002699
C-A    -14 -16.548741 -11.451259 0.0000003
C-B     -8 -10.548741 -5.451259 0.0000283
```

2. Effect of Irrigation at each Fertilizer

```
model_A <- aov(Yield ~ Irrigation, data = subset(data, Fertilizer == "A"))
TukeyHSD(model_A)
```

Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = Yield ~ Irrigation, data = subset(data, Fertilizer == "A"))

```
$Irrigation
      diff      lwr      upr    p adj
Low-High  -16 -18.23371 -13.76629 2.1e-06
```